

PRANAY KABRA

Manufacturing Engineering Graduate | (236) 999-0388 | pranaykabra13@gmail.com | [LinkedIn](#) | [Website](#)

EDUCATION

University of British Columbia

Bachelor of Applied Science, Manufacturing Engineering, Minor in Commerce

Vancouver, BC

Expected May 2026

- Key Courses: Industrial Robotics & Automation, Computer Control of Mechatronics Systems, AI and ML in Manufacturing

WORK EXPERIENCE

Svante

Mechanical and Material Engineering Co-op

Vancouver, BC

January 2025 to August 2025

- Designed and fabricated a **GD&T**-controlled lap shear testing jig to support adhesive strength validation, reducing sample preparation time by **40%** and improving test repeatability
- Enhanced test bench stability by designing custom structural supports in **Creo** and implementing a modular layout using off-the-shelf components, reducing measurement variability and cutting product changeover time by **60%**
- Implemented operator safety systems for commissioning a production-ready cobot assembly line by integrating a **PILZ** motion radar sensor, developing a **state-machine architecture in Python**, and creating an **HMI** for real-time monitoring
- Calibrated a hot-wire anemometer for near-field airflow measurement by integrating reference velocity and pressure sensors; programmed a **Beckhoff PLC** and developed an **HMI** interface to record velocity-voltage data
- Automated test data analysis by using **VBA** and macros in Excel, reducing report generation time by **83%**
- Collaborated cross-functionally with design, quality, and manufacturing engineers to improve product manufacturability.

Dometic Marine Solutions

Quality Engineering Co-op

Vancouver, BC

September 2024 to December 2024

- **Saved \$200,000** in scrap costs by bridging product manufacturing and quality inspection; applied Lean principles with time studies, root cause analysis, and value stream mapping to cut scrap rate and improve productivity
- Led continuous improvement initiatives and reduced downtime from **3 days to 7 minutes** by optimizing material flow and eliminating non-value-added motion
- Performed Gauge Repeatability & Reproducibility study and provided recommendations to improve measurement accuracy

TECHNICAL EXPERIENCE & PROJECTS

Capstan Gearbox Development

UBC Capstone Project

Vancouver, BC

September 2025 to Present

- Led the development of a capstan drive to replace a conventional gearbox in a knee orthotic device, eliminating backlash, reducing weight, and improving torque transmission within strict packaging constraints
- Designed and incorporated a **linear actuator** to maintain cord tension and enable repeatable re-tensioning over time
- Developed torque and endurance test benches using an **Arduino Nano** with stepper motor and driver circuitry
- **Machined** custom aluminum components on a lathe and mill after improving part strength to support continued validation

UBC Smart City, Design Team

Subteam Lead

Vancouver, BC

September 2024 to Present

- Managed a 10-member multidisciplinary team to develop a net-zero, wind-turbine-powered smart streetlight
- Modeled wind turbine blades in SolidWorks and fabricated Al 5052 prototypes via waterjet cutting and sheet metal bending
- Designed a 3-stage planetary gearbox in **Fusion 360** achieving 300× RPM multiplication for low-wind-speed operation

RC Car Component Design & Manufacturing

Design Lead

Vancouver, BC

September 2023 to April 2024

- Designed wheels and shell model in Fusion 360, **3D printed** the shell, and fabricated the shell using thermoforming
- Performed tensile and 3-point bending test to guide chassis and shell material selection

SKILLS

- **CAD and Computer Software:** Siemens NX, GD&T, Creo, SolidWorks, Fusion 360, Vention, ARAS PLM, Microsoft Office
- **Programming and Simulation:** Python, Ladder Logic, Structured Text, MATLAB, VBA, ANSYS, Arena, ABAQUS
- **Manufacturing and Technologies:** CI, 5S, FMEA, DFMA, GR&R, 3D Printing, Arduino, Beckhoff, HMI Design, ESP32